Data Analytics on Accident Data for Smarter Cities and Safer Lives

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Abstract— **Analysis of accidents has always been very important to take major decisions. In this paper, analysis of accident data set is done, where the dataset consists of the information related to accidents such as the type of accidents, condition of light during the accident, severity of accident, speed zone, consumption of alcohol involved in the accident etc. The focus is laid on the application of data analytics to predict and reduce the impact of the accidents and uncover the relevant trends by training the system with the past data using decision tree methodology. The major causes leading to the accidents are identified and analyzed using some of the relevant attributes such as alcohol consumption, hit and run cases etc. Some demographic graphs are also plotted to bring out better visual insights.**

Keywords-Accidents, Decision Trees, Data Analytics, R Analysis, Pruned model

I. INTRODUCTION

The survey by NSW in [1] shows that from 2005 to 2013, about 30 people are hospitalized by crashes on roads each day and UN reports in [2] that 30,000 people die every year due to the road accidents. Researchers use common data mining and prediction models on the wealth of accident related information available. The intent of this paper is to examine and ponder on the solution to accidents and accident response in urban areas. Modules have been created to deal with the huge datasets and to bring out unique insights into the results by doing accident data analysis [3]. The module involving the determination of the severity of the accidents, on the basis of certain factors such as the city, consumption of alcohol, speed zone, the condition of the light etc. The above prediction is done using Decision Tree analysis [4,5] which generates a decision tree as a result of prediction.

II. FRAMEWORK

The input to the process is a huge dataset is in .csv format. The R system is used to perform normalization [5], pre-processing, distribution and other major analytic data steps. Data Cleaning [4,7] is done to obtain the proper data for analysis purpose. The data is transformed into a format that the analysis and implementation of algorithm becomes simpler. We then prepared a hypothetical analytic model plan using sample data, which is referred to as the testing data in the system architecture. Some useful inferences are also drawn from the results which are obtained from the implementation on the testing data.

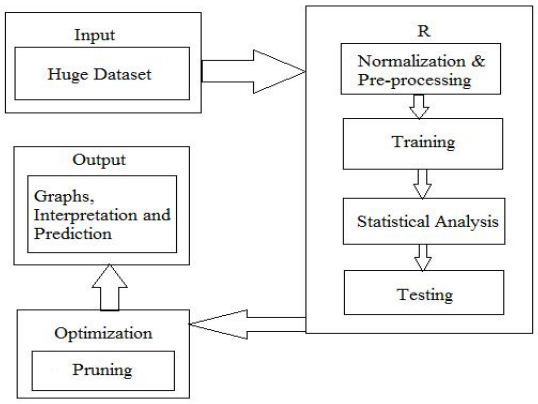

Figure 1.Design of the complete system

The statistical analysis as in [8] is first performed on the training data and then on the testing data. The deviation of the behavior of the testing data from the training data is also measured. An optimization, called Pruning as in [9] is performed later in order to minimize the deviation of the testing data from the training data. This deviation is measured in the terms of the mean values of all the differences. The lesser is this value, the better is the performance of the analysis as lesser is the errors which are possible. After the pruning of decision trees is performed, the optimal prediction and the decision tree is also generated, which gives us an easier as well as faster way to understand the process of reaching at the conclusion about the severity of accidents.

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III. IMPLEMENTATION

The dataset has been received by the Office of the Traffic Commissioner at Bangalore. It consists of the following columns and has the possible values as mentioned below:

TABLE I. DESCRIPTION OF THE DATASET

Column Names	Description	Possible values and their meanings
Accident ID	Uniquely identifies the accident tuples present in the dataset.	Unique integers
Accident Type	Describes the cause of accident	1-Consumption of Alcohol 2-Collision with vehicles 3-Absence of Traffic lights 4-Struck pedestrian 5-Collision with animals 6-Falling in ghats
LIGHT_CONDITION	It tells the phase of the day when the accident took place	1-Night 2-Dusk 3-Dawn 4-Noon
Severity	It denotes the severity of the accident	0-Low 1-Moderate 2-High
Speed Zone	It tells the speed zone of the area where the accident occurred.	It has values of speeds from 30 to 100 km/hr.
ALCOHOL_TIME	It tells whether alcohol was consumed or not.	0-Yes 1-No
Pedestrian Count	It shows the number of pedestrians who were injured due to the accident.	Integer values
No. of Injured people	It shows the total number of people who were injured due to the accident.	Integer values

The reading of data, normalization as in [10] of data, training the machine, implementation of the algorithms, statistical analysis, testing of the accuracy of the predicted result, further optimization etc. is done using the analytical tool, R. It has exhaustive libraries which support the prediction, analysis and implementation of the algorithms. CRAN stands for Comprehensive R Archive Network [11]. It is the collection of many sites which contain similar kind of material, consisting of the R distribution, extensions, documentations and the binaries.

The packages used for the implementation are:

a) Plotrix: This package is used for a large number of specialized plots and functions like color scaling, text placement and legends. It allows users to get many sorts of specialized plots quickly. It also allows easy customization of those plots without learning a great deal of specialized syntax.
b) Tree: It runs a K-fold cross-validation experiment as in [5] to find the deviance or number of misclassifications as a function of the cost-complexity parameter k. The overall deviance or a vector of contributions from the cases at each node. The overall deviance is the sum over all the leaves in the decision tree in the latter case.
c) ggplot2**:** It is an implementation of the grammar of graphics in R. It combines the advantages of lattice and base graphics conditioning as in and the shared axes are maintained by it. It also implements a sophisticated multidimensional conditioning system and a consistent interface to map data to aesthetic attributes.
d) ISLR: It is the most commonly used package to plot the decision trees [12] based on certain parameters and to prune the tree based on the tree generated leading to the better accuracy of the prediction.

Decision tree builds classification as in [1,3,8] or regression models [7] in the form of a tree structure. It breaks down dataset into smaller subsets while at the same time an associated decision tree is developed incrementally. The final outcome is a tree with the decision nodes at different levels and the leaves as the results denoted by leaves in the tree.

This algorithm uses the concept of entropy to find out the homogeneity of the data sample. When the sample is absolutely homogeneous, then the entropy will be equal to zero. If the sample is equally partitioned, it has the entropy value equal to 1. The dataset is then split based on the different attributes and some of the based conditions The entropy [14] of the data sample is obtained using the formula, where H(S) is the entropy as in [5] and p(x) is the number of successful cases. X is the set of all the events and x is one event out of those, used one at a time.

$$H(S) = - \sum_{x \epsilon X} p(x) log_2 p(x) \quad (1)$$

The dataset is then split based on the different attributes. The entropy for each branch is calculated. Then it is added proportionally, to get total entropy for the split. The resulting entropy is subtracted from the entropy before the split. The result is the Information Gain [5], or decrease in entropy. This is obtained using the formula below, where IG is the Information Gain, H is the entropy and p is the number of successful cases which are present for each case.

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$$IG(A, S) = H(S) - \sum_{t \epsilon T} p(t) H(t) \qquad (2)$$

The attribute with the largest information gain was chosen as the decision node as in [15]. A branch with entropy of 0 is a leaf node. The branches with value of entropy greater than 0 need further splitting. The algorithm runs recursively on the non-leaf branches, until the data gets classified. The final decision tree gets generated at the last.

IV. MODULES

A. *Analysis of Severity of Accidents based on consumption of alcohol*

In this module, the severity of the accidents based on alcohol consumption is examined. The values on the X-axis correspond to the severity of the accidents being highly, moderately and least severe. The Y-axis shows the number of accidents which have occurred due to the consumption and non-consumption of alcohol represented. The conclusion which is deduced from this is that most of the cases when the driver of the vehicle has consumed alcohol, the severity was more, which would definitely have caused the accident. The number of severe accidents which have occurred due to the consumption of alcohol is nearly twice the number of severe accidents occurred due to the non-consumption of alcohol. Even the accidents which were moderately severe are quite good in number. There are also many accidents which have occurred due to the non-consumption of alcohol. Figure 2 shows the bar graph for this analysis.

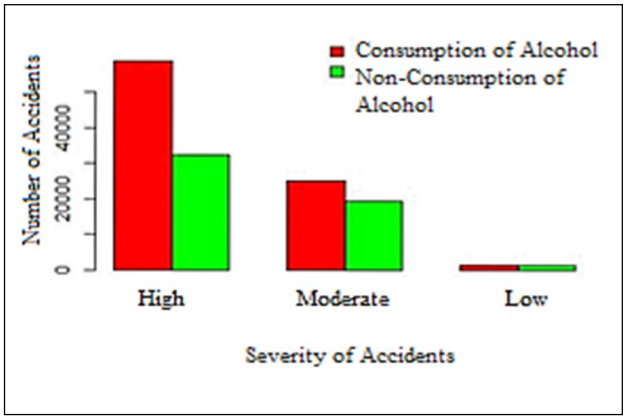

Figure 2.Analysis of Severity of Accidents based on alcohol consumption

The pseudo code for this module is as below:

```
Step 1: Import the libraries
Step 2: Extract the dataset in R variable
        mydata=read.csv("Source.csv")
Step 3: Plot the graph
      table_data <-table(mydata$SEVERITY,
      mydata$ALCOHOL_CONSMUPTION)
      barplot(table_data, xlab = "Severity of Accidents",ylab="
      Number of Accidents",col=c("red","green"),
      legend=rownames(table_data),beside = TRUE)
```

B. *Analysis of Severity of Accidents based on the phase of the day*

In this module, the relationship between the severity of accidents and the phase of day is identified. The X-axis shows the various light conditions such as complete darkness with no street lights, dawn, dusk and afternoon. The accidents were extremely higher in number and were more severe, when there was pitch-darkness and street lights were absent (greater than 60,000 in number). The number of highly severe accidents caused during the dawn time (close to 5,000) is more than the number of severe accidents occurred during the dusk and afternoon. The number of moderate accidents occurred during the dawn and dusk are very much considerable too as they are large in number. The least, moderately and highly severe accidents mostly don't occur during the noon. Figure 3 below shows the bar graph for analysis as discussed in this module.

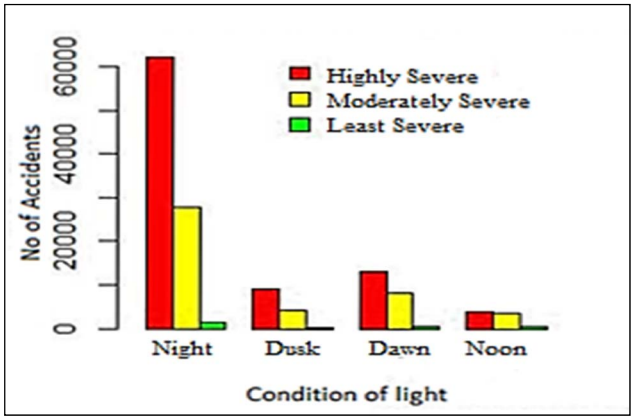

Figure 3.Analysis of severity of Accidents based on the light conditions or the phase of day

The pseudo code for this module is as below:

```
Step 1: Import the libraries
Step 2: Extract the dataset in R variable
        mydata=read.csv("Source.csv")
Step 3: Plot the graph
      table_data <-table(mydata$LIGHT_CONDITION,
      mydata$NOA)
      barplot(table_data, xlab = "Condition of
      light",ylab="No of Accidents", Col=c("Red",
          "yellow", "green", legend=rownames(table_data))
```

C. *Analysis of Severity of the Accidents based on the speed zones*

The X-axis corresponds to the severity of the accidents being highly, moderately and least severe. The majority of severe accidents occur in the speed zones where allowable driving

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speed is 60 km/hr. The reason could be too many vehicles moving in those zones causing mayhem and commotion in the dense roads. The number of accidents happened in the zones where allowable speed is nearly 50 and 100 km/hr is also pretty much considerable. The majority of moderately severe accidents occur in the zones where the driving limit is 70 km/hr, followed by the places with a speed limit of 40 km/hr.

One important inference from this is that maximum accidents don't occur in the high speed zones but they are in the average speed zones only. The number of less severe accidents is comparatively quite less in number and have mostly occurred in the areas with a speed limit of 50km/hr. Overall, the accidents are very unevenly distributed here. The figure 4 shows the results for this analysis.

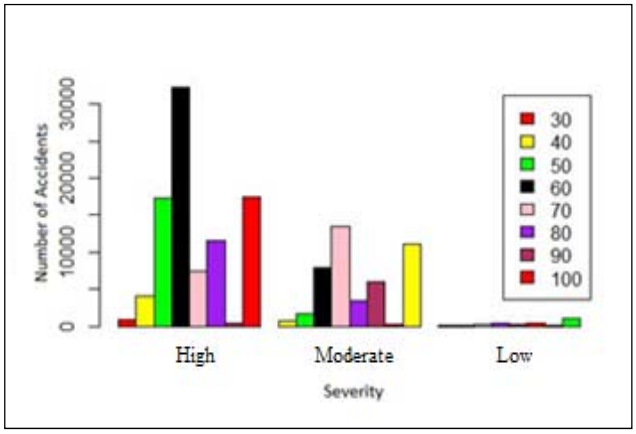

Figure 4.Analysis of Severity based on the speed zones

The pseudo code for this module is as below:

Step 1: Import the libraries
Step 2: Extract the dataset in R variable
mydata=read.csv("Source.csv")
Step 3: Plot the graph
table_data <-table(mydata$SEVERITY,mydata$NOA)
barplot(table_data, xlab = "SEVERITY",ylab="
No of Accidents",col=c("red","yellow","green",
"black","pink","purple","violet"),
legend=rownames(table_data),beside = TRUE)

D. Decision tree to find the severity of the accident using consumption of alcohol and light condition as factors

In this module, the decision tree algorithm as in [1,3] was implemented to discern the cause of accidents in the metropolitan areas. The alcohol time was used as the root node, if the alcohol is consumed and there is any pedestrian on the road, the severity of accident becomes high, if not, it is low. The right side corresponds to the non consumption of alcohol.

If the condition of light is 0, which means that it is total darkness and if more than one person has got injured, then the severity is high; in the other case it has low severity. If the light condition is greater than 0, which means dawn, dusk and afternoon, the accidents are moderately severe. The decision tree for this is shown in figure 5.

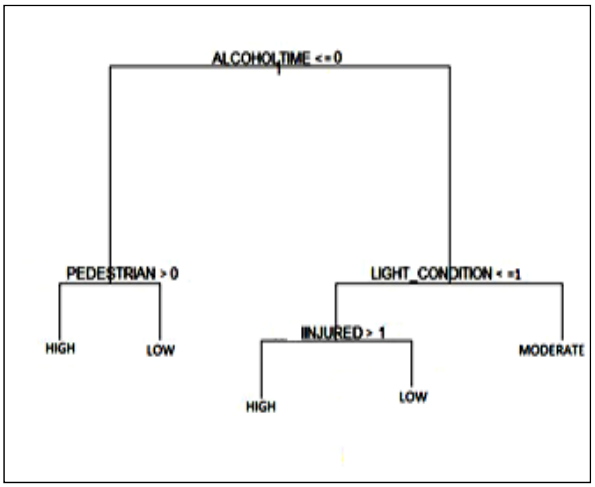

Figure 5.Decision Tree to find the severity of the accident using consumption of alcohol and light condition

The steps as in the pseudo code for the implementation of the decision trees are mentioned below:

1. Import all the required libraries required for the analysis.
2. Clean the dataset and extract the entire dataset in a R variable.
3. Identify the dependant and independent variables present in the dataset.
4. Analyze the relation between dependent and independent variable based on certain conditions.
5. Attach the new observation column corresponding to the results obtained in Step 4 to the dataset.
6. Remove the dependant column from the extracted dataset.
7. Apply the decision making algorithm, ID3 on the training data sample as in [16].
8. Apply decision making algorithm, ID3 on the testing data sample.
9. Measure the accuracy of the prediction based on the results obtained from Step 7 and Step 8.
10. Optimize the prediction by performing the prune the decision tree obtained from Step 9. Repeat Step 9 until the value of the errors in prediction becomes the least.

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V. OPTIMISATION AND PERFORMANCE

The decision tree algorithm makes use of entropy [14] which is calculated using a formula, specified earlier. Using the value of the entropy, the largest information gain value is found. The information gain as in [5] represents nothing but the extent to which the prediction of the event happens correctly using the method or algorithm used for training the machine. The comparison of the values of the means of the deviations for the module 4.4 is shown in the table below. The value of the mean of the deviations decreased from 0.4337385 to 0.4247297 as shown in Figure 6, after performing pruning which is described in [6,12] for the optimization.

TABLE II.COMPARISON OF MEAN OF DEVIATIONS BEFORE AND AFTER PRUNING

Is Pruning Performed?	Mean of Deviations
No	0. 4337385
Yes	0. 4247297

```
> plot(tree_model)
> text(tree_model,pretty=0)
> test_pred=predict(tree_model,testing_data,type="class")
> testing_high=high[test]
> mean(test_pred!=testing_high)
[1] 0.4337835

> plot(cv_tree$size,cv_tree$dev,type="b")
> pruned_model=prune.misclass(tree_model,best=2)
> text(pruned_model,pretty=0)
> tree_pred=predict(pruned_model,testing_data,type="class")
> mean(tree_pred!=testing_high)
[1] 0.4247297
```

Figure 6.Comparison of means of the deviations before and after pruning

Initially, when there was no pruning [9] introduced to the decision tree, the mean of the deviations was almost 0.433. The pruning basically shows us a graph using which the numbers of levels to be used for the decision making can be decided. Usually, the number of levels for the pruning lies in the middle of the range. Here, the value of 2 is used for the pruning, which means that it would generate a new tree based on the new level value.

VI. COMPARISON WITH THE EXISTING WORK

The existing analysis of traffic data focuses on the accident statistics, causes of accidents, data collection, accident reconstruction, safety measures and safety audit. This paper aims to uncover the patterns followed by the accidents. The phase of day when most of the highly severe accidents occur is found. The relationship between the speed zones where most of the accidents occur and their severity is also found out. Our analysis lightens the hidden causes so that the number of accidents could be minimized by taking preventive steps. The financial losses could also be reduced. The idea of using decision trees on the accident data helps to find the causes which helps us to identify the various factors which lead to highly, moderately and least severe accidents. The primary causes of accidents are also identified. The number of pedestrian and people injured in the accidents are used to predict the severity of the accidents.

VII. CONCLUSION

The analysis of the accident data is performed quantitatively and qualitatively. The sequences of conditions which lead to severe accidents are examined using the decision trees. Also, regions of high density accidents, the severity of the accident and the attributing factors were examined. The project in its early iteration can glean insight into locations which are accident prone and the causes for it. This predictive model can be implemented in several mobile applications which can minimize risk of accidents. It also attempts to research the possible causes of accident and the possible solutions to them.

VIII. FUTURE WORK

This system aims to improve the approach of solving the method of collection and analysis of accident related information. The collection of real time data can also be done by placing vibration or shock sensors, tilt sensors, fire sensors etc. inside the vehicle. GPS Module can be used to obtain the precise latitude and longitude which can be sent to the nearest police station as well as the hospital based on the severity of the accident to ensure less time spent in arrival of the police or ambulance to the accident spot. The cloud infrastructure can be used to store this data. This data can be used to implement protective measure to ensure road-safety of elderly citizens, wildlife and school children etc. This data can also be used to build hospitals in locations so as to optimize response time and ensure timely arrival of ambulances, provide cab service to the people when they have consumed alcohol etc. These results can also be used by the policy-makers to design policies which ensure more and better road safety.

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